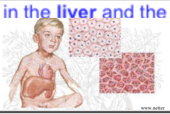
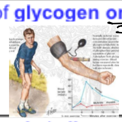
GSD Type Organ	Deficient enzyme	Glycogen amount and structure	Clinical features Blood glucose levels
I Von Gierke disease Liver and kidney	Glucose 6-phosphatase in ER or glucose 6-P translocase (rare)	Increased glycogen amount in the liver and the kidney. Normal glycogen structure. 	Massive enlargement of liver and kidney disease. AVOID FASTING METABOLISM Severe hypoglycemia, ketosis, lactic acidemia, hyperuricemia, hyperlipidemia. Early death if untreated. Failure to thrive. Treatment at night: intake of uncooked corn starch or nocturnal nasogastric glucose infusion. Txt: uncooked cornstarch q few hrs @ night OR NOCTURNAL NASOGASTRIC GLUCOSE INFUSIONS
II Pompe disease All organs Lysosomes	Lysosomal (acid) α-1,4-glucosidase (acid maltase) 	Massive increase in amount of glycogen in lysosomes with normal glycogen structure mainly in heart, liver and muscle. Danger of lysosomal rupturing in severe cases. (glycogen storage, lysosomal storage, & mscl dz)	Infantile form: Massive cardiomegaly and hypotonia (floppy) and muscle weakness. Elevated serum CK-MM. If untreated leads to early death by age 2 years due to cardiorespiratory insufficiency. Less severe late-onset forms lead to respiratory weakness, myopathy or muscular dystrophy in later life. Treatable with enzyme infusion. Normal blood glucose levels. early death. myopathy, progressive neuromsc disorder
III Cori disease Forbes disease Liver and muscle, heart (limit dextrinosis)	Glycogen debranching enzyme (4:4 transferase or α-1,6-glucosidase) 	Increased amount of glycogen in liver, muscle and heart. Large glycogen particles. Abnormal glycogen structure with short branches (limit dextrinosis). (an intermediate structure accumulates)	Mild hypoglycemia and hepatomegaly in children. Muscle, heart: weakness, hypotonia and cardiomyopathy. Muscular dystrophy in later life. Treatment with frequent meals to avoid fasting, diet high in protein. -mild hypoglycemia -a type of DYSTROPHY -Pt's may be in wheelchair by 50-60 y/o
IV Andersen disease Liver and muscle	Glycogen branching enzyme (4:6 transferase)	Low amount of glycogen in liver and muscle. Abnormal glycogen structure with long → hepatic scarring unbranched glucose chains (attack by immune system causing scarring).	Enlarged liver and spleen, hepatomegaly due to infantile cirrhosis. (early death) Hypotonia, myoglobin in urine (burgundy colored). Early death. msc weakness/weakened lungs Hypoglycemia when cirrhosis occurs. -large variety & severity
V McArdle disease Muscle	Muscle glycogen phosphorylase MYOPHOSPHORYLAZE ENZ	Moderately increased amount of glycogen only in muscle. Normal glycogen structure. 	During exercise: Muscle weakness and cramping due to lack of ATP. (burgundy colored urine) following possible rhabdomyolysis. No increase of lactate in blood after exercise as glycogen degradation is deficient. Patient is normal well developed. -babies not dx b/c child just appears tired Normal blood glucose levels.
VI Hers disease Liver	Liver (HEPATOPHSPHRLYASE) glycogen phosphorylase	Increased amount of glycogen only in liver. Normal glycogen structure.	Liver: mild hypoglycemia. Hepatomegaly, growth retardation.
VII Tarui disease Muscle and RBC	Glycolysis: Muscle PFK-1 RBC PFK-1 (M isoform)	Increased glycogen amount in muscle. Normal glycogen structure. Deficient glycolysis in muscle and RBC.	Muscle: lack of ATP leads to muscle cramping, no rise in lactate. RBC: lack of ATP leads to hemolysis. (due to 50% PFK-1 deficiency in RBC) Normal blood glucose levels. -similar to McArdle but SOM.1a.BPM2.3.DM.2.BCHM.MB.0906

Very Pompous Couriers Add McChickens & Hershey's Kisses To my order
GbPase Lysosomal α1,4 debranching branching Myophos.ase Hepatophos.ase PFK-1

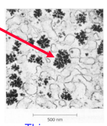
Type 0 = glycogen synthase deficiency (rare, early death)

The largest amount of glycogen is found in skeletal muscle (about 400 g).

The highest concentration of glycogen is found in liver. (about 100 g, up to 10% of fresh weight).

The branched structure of glycogen allows rapid synthesis after a meal or rapid degradation when needed at the nonreducing ends.
Branches are formed by α-1,6 glucosidic bonds.

Glycogen is stored in cytosolic glycogen granules mostly in the liver and muscle.



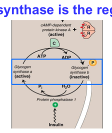
The enzymes for glycogen synthesis and for glycogen degradation are in the cytosol close to the glycogen granules and are tightly controlled in their activities.

A small amount of glycogen is degraded in lysosomes. This was discovered in patients with Pompe Disease where lysosomal degradation is deficient and glycogen accumulates in lysosomes.

The purpose of glycogen synthesis in muscle and liver is storage of glucose when the blood glucose level is high.

Glycogen synthase is the regulated enzyme.

Insulin leads to activation and dephosphorylation by protein phosphatase 1.



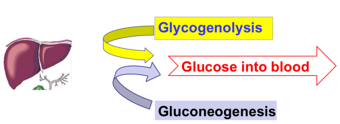
Glucagon and epinephrine lead to inactivation and phosphorylation by protein kinase A (cAMP).

LIVER glycogen degradation has the purpose to provide glucose 6-P for release of free glucose into the blood.

MUSCLE glycogen degradation has the purpose to provide glucose 6-P for muscle glycolysis and energy metabolism.

Liver glycogen degradation	Muscle glycogen degradation
needs always hormonal activation mostly by glucagon and epinephrine. Glycogenolysis shall take place only during fasting and stress situations for release of glucose into the blood for other tissues.	is activated by calcium and AMP during muscle contraction independent of hormones. In flight and fight situations: epinephrine accelerates muscle glycogen degradation.

SOM.1a.BPM2.3.DM.2.BCHM.MB.0



Glycogen degradation and gluconeogenesis take place at the same time during fasting and low blood glucose. Both pathways are under tight regulation by hormones.

Liver metabolism after action of glucagon and epinephrine

Glycogen synthesis and glycolysis are inhibited.
Glycogen degradation and gluconeogenesis are active.

cAMP-dependent protein kinase A phosphorylates directly

- Glycogen synthase P (Glycogen synthesis inhibited)
- Glycogen phosphorylase kinase P (Glycogen degradation activated) → Glycogen phosphorylase P (activate)
- Pyruvate kinase P (Glycolysis inhibited)
- P Bifunctional enzyme (Gluconeogenesis activated)

Insulin leads to dephosphorylated key enzymes

- Glycogen synthase: activated glycogen synthesis.
- Glycogen phosphorylase kinase: inhibited glycogen degradation.
- Bifunctional enzyme: activated glycolysis and inhibited gluconeogenesis.
- Pyruvate kinase: activated glycolysis.

Glucagon leads to phosphorylated key enzymes

- Glycogen synthase-P: inhibited glycogen synthesis.
- Glycogen phosphorylase kinase-P: activated glycogen degradation.
- Bifunctional enzyme-P: activated gluconeogenesis and inhibited glycolysis.
- Pyruvate kinase-P: inhibited glycolysis and PEP saved for gluconeogenesis.

High insulin, low glucagon Fed state: GLYCOGENESIS and GLYCOLYSIS	Low insulin, high glucagon Fasting: GLYCOGENOLYSIS and GLUCONEOGENESIS
At high blood glucose: uptake of glucose into liver	At low blood glucose: release of glucose into blood
GLUCOSE ↓	GLUCOSE ↑
Glucokinase	Glucose 6-phosphatase
Phosphofructokinase-1	Fructose 1,6-bisphosphatase
Bifunctional enzyme: is dephosphorylated Active PFK-2	Bifunctional enzyme is phosphorylated: Active BPase-2
Pyruvate kinase dephosphorylated	PEP carboxykinase Pyruvate carboxylase
Continued in PDH and TCA cycle	β-oxidation is required

SOM.1a.BPM2.3.DM.2.BCHM.MB.09